

Electrical Circuits I

ENGR 2405

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Chapter 11 – AC Power Analysis

Chapter 12 – Three-Phase Circuits

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Overview

- AC power will be covered in this module
- The differences between instantaneous v average, resistive v reactive, and apparent v complex powers will be covered
- Three phase AC power will be introduced along with the four configurations for sources and loads will be discussed

Instantaneous Power

- This is the power at any instant in time
- It is the rate at which an element absorbs power
- Consider the generalized case where the voltage and current at the terminals of a circuit are:

$$v(t) = V_m \cos(\omega t + \theta_v) \quad i(t) = I_m \cos(\omega t + \theta_i)$$

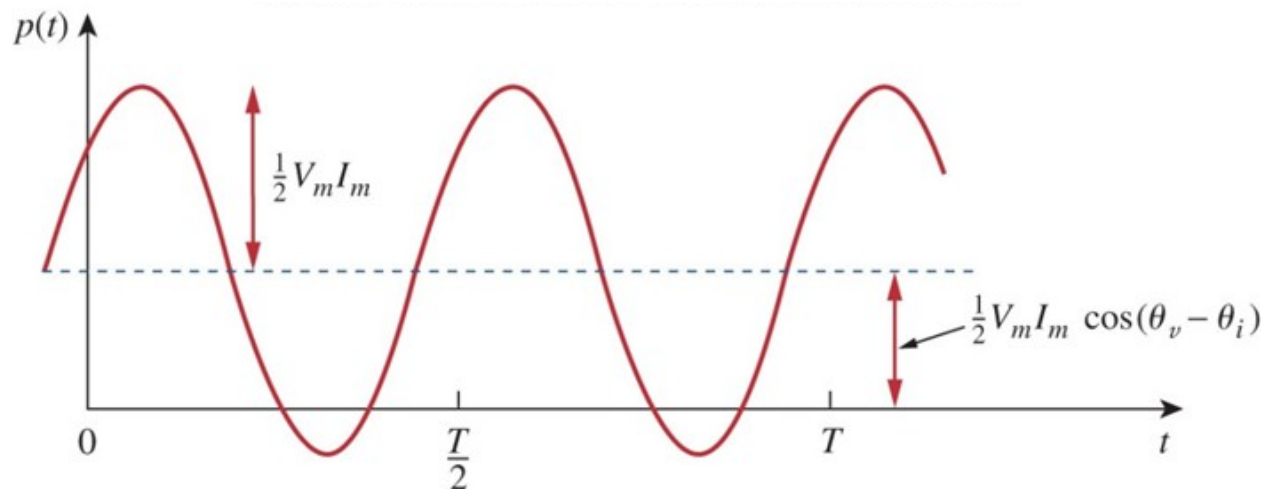
- Multiplying the two together, yields:

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Instantaneous Power

There are two components:

- One is constant, depending on the phase difference between the voltage and current
- The second is sinusoidal with a frequency twice that of the voltage and current



Instantaneous Power

- Note that the power can go negative!
- This is due to circuit elements which can store and release energy (e.g. caps)
- Note also that instantaneous power is constantly changing => hard to measure
- Thus, it's more common to measure average power

Average Power

- Average power is the instantaneous power averaged over a period, given by:

$$P = \frac{1}{T} \int_0^T p(t) dt$$

- The time dependent part of instantaneous power averages is a sinusoid $\Rightarrow$ averages to 0, thus this returns the part that is constant
- With instantaneous power, you have to work in the time domain, but with average power, we can be work in frequency domain where it is:

$$P = \frac{1}{2} \Re(VI^*) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Resistive versus Reactive

- When $\theta_v = \theta_i$ the voltage and current are in phase and the circuit is purely resistive:

$$P = \frac{1}{2} V_m I_m = \frac{1}{2} I_m^2 R = \frac{1}{2} |I|^2 R$$

- When $\theta_v - \theta_i = \pm 90^\circ$, the voltage and current are 90° out of phase (circuit absorbs no power) and is purely reactive:

$$P = \frac{1}{2} V_m I_m \cos(90^\circ) = 0$$

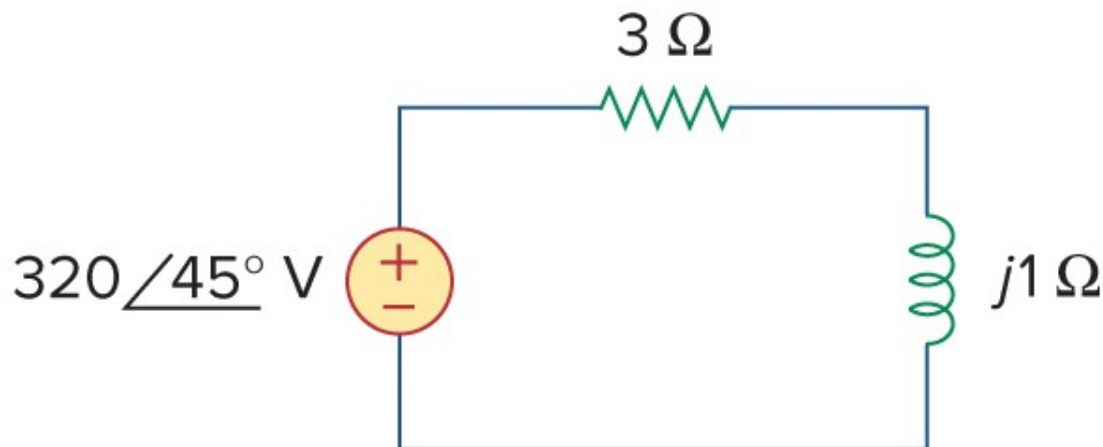
Practice Problems

11.1 Calculate the instantaneous power if:

$$v(t) = 330 \cos(10t + 20^\circ) \text{ V}$$

$$i(t) = 33 \sin(10t + 60^\circ) \text{ A}$$

11.3 Calculate average power absorbed by R and L and find average power supplied in:



Maximum Average Power Transfer

- For DC, max power to a resistive load which occurs when the load is equal to the Thevenin resistance
- In AC, since we're dealing with complex numbers:
- Thevenin impedance/load: $Z = R + jX$
- Current through load:

$$I = \frac{V_{Th}}{Z_{Th} + Z_L} = \frac{V_{Th}}{(R_{Th} + jX_{Th}) + (R_L + jX_L)}$$

- Average Power delivered:

$$P = \frac{1}{2} |I|^2 R_L = \frac{|V_{Th}|^2 R_L / 2}{(R_{Th} + jX_{Th})^2 + (R_L + jX_L)^2}$$

Maximum Average Power

- We want to find the peak in this function => take derivative and set it to 0 for both R_L and X_L :

$$\frac{\partial P}{\partial X_L} = - \frac{|V_{Th}|^2 R_L (X_{Th} + X_L)}{\left[(R_{Th} + jX_{Th})^2 + (R_L + jX_L)^2 \right]^2}$$

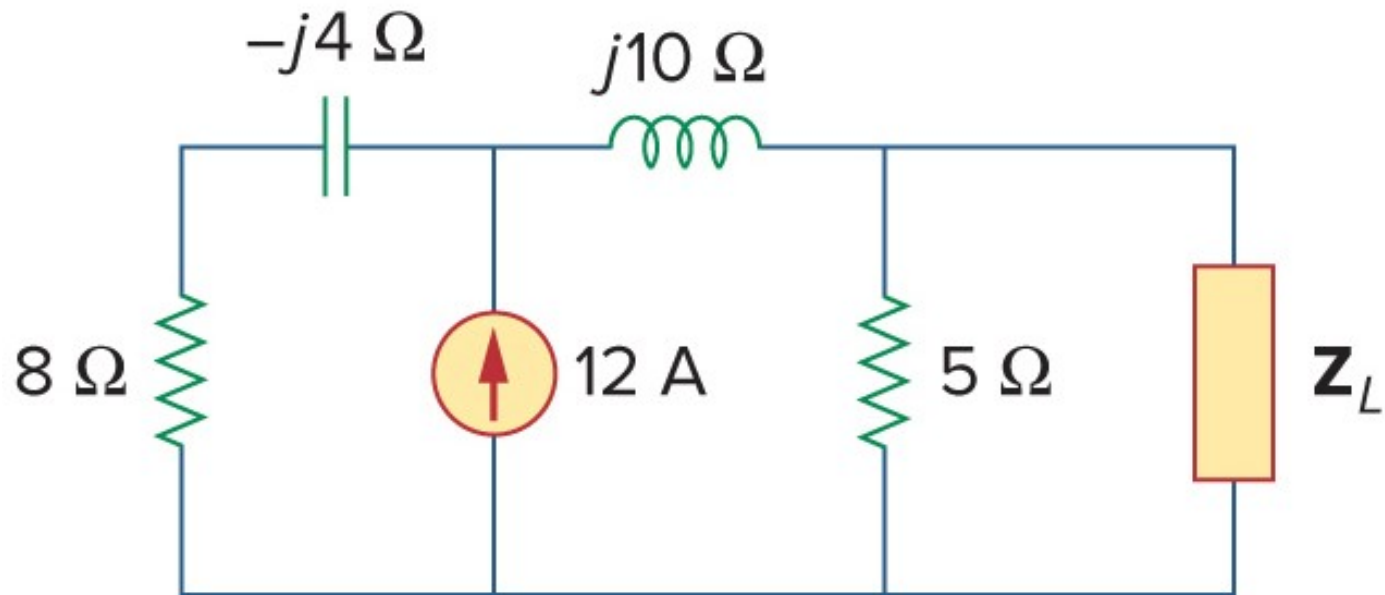
$$\frac{\partial P}{\partial R_L} = - \frac{|V_{Th}|^2 (R_{Th} + R_L)^2 + (X_{Th} + X_L)^2 - 2 R_L (R_{Th} + R_L)}{\left[(R_{Th} + jX_{Th})^2 + (R_L + jX_L)^2 \right]^2}$$

- which can be satisfied if $Z_L = Z_{Th}^*$ resulting in:

$$P_{max} = \frac{|V_{Th}|^2}{8 R_{Th}}$$

Practice Problems

11.5 Find $\mathbf{Z}_L$ for maximum power transfer in:



Effective Value

- A DC current delivering the same average power to a resistor as an AC current is called the effective current
- For a periodic current, the average power absorbed is:

$$P = \frac{1}{T} \int_0^T p \, dt = \frac{R}{T} \int_0^T i^2 \, dt = i_{eff}^2 R \Rightarrow i_{eff} = \sqrt{\frac{1}{T} \int_0^T i^2 \, dt}$$

- Similarly

$$v_{eff} = \sqrt{\frac{1}{T} \int_0^T v^2 \, dt}$$

- These are also known as their RMS (root mean square) values and works with any periodic value

RMS

- If the parameter of interest is a constant, the RMS value will be that constant
- For a sinusoidal waveform, the RMS value is related to the amplitude as follows:

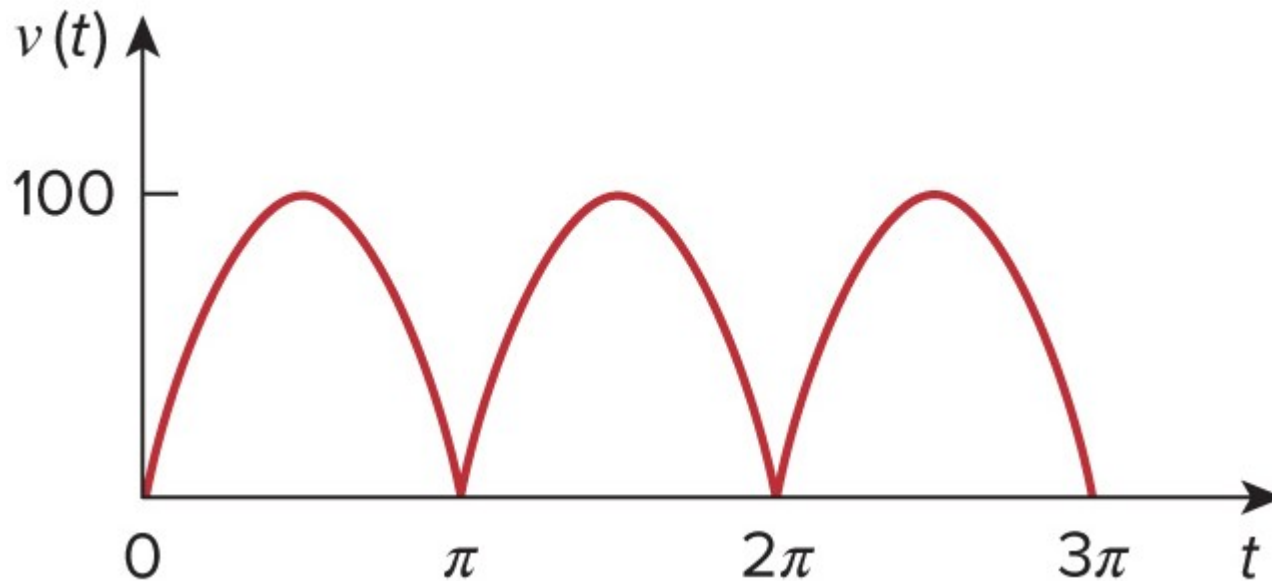
$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

- RMS power can be determined from V_{rms} or I_{rms} :

$$P = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

Practice Problems

11.8 Find rms value of the following sine wave and calculate average power dissipated by 6Ω resistor



Apparent Power

- Multiplying V_{rms} by I_{rms} : is called "apparent" power
- It is modulated by the phase difference:
 $\cos (\theta_v - \theta_i)$
- which is known as the "power factor" (with the angle difference being called the pf angle)
- It is called "apparent" power because that is what the power in the circuit seems to be (but isn't)
- It is measured in VA, to distinguish it from the average or real power

Power Factor

- The power factor can range from 0 to 1
- If the pf angle is positive, the current leads the voltage, and vice versa
- Purely reactive loads will have a power factor of zero
- Power factors affect the way utilities bill for electricity

Practice Problems

11.9 Obtain the power factor and apparent power of a load whose impedance is:

$$Z = 60 + j40 \, \Omega$$

when the applied voltage is:

$$v(t) = 320 \cos(377t + 10^\circ) \, V$$

Complex, Real, & Reactive Power

- Complex power is the total effect of parallel loads on power. In rectangular form:

$$S = I_{rms}^2 (R + jX) = P + jQ$$

- P is real power and Q is "reactive" power
- Real power is the only useful power and is measured in Watts
- Reactive power is a measure of the energy exchange between the source and the reactive load (e.g. inductor or cap) and is measured in units of volt-ampere reactive (VAR)

Summarizing Power

$$\text{Complex Power } = S = P + jQ = V_{rms} (I_{rms})^*$$

$$= |V_{rms}| |I_{rms}| \angle (\theta_v - \theta_i)$$

$$\text{Apparent Power } = S = |S| = |V_{rms}| |I_{rms}| = \sqrt{P^2 + Q^2}$$

$$\text{Real Power } = P = \text{Re}(S) = S \cos(\theta_v - \theta_i)$$

$$\text{Reactive Power } = Q = \text{Im}(S) = S \sin(\theta_v - \theta_i)$$

$$\text{Power Factor } = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

Practice Problems

11.11 For a load $V_{rms} = 110 \angle 85^\circ V$ and $I_{rms} = 400 \angle 15^\circ mA$

Determine:

- a. Complex and apparent powers
- b. Real and reactive powers
- c. Power factor and load impedance

11.12 A sinusoid source supplies 10kVAR reactive power to a load $Z = 250 \angle -75^\circ \Omega$

Determine:

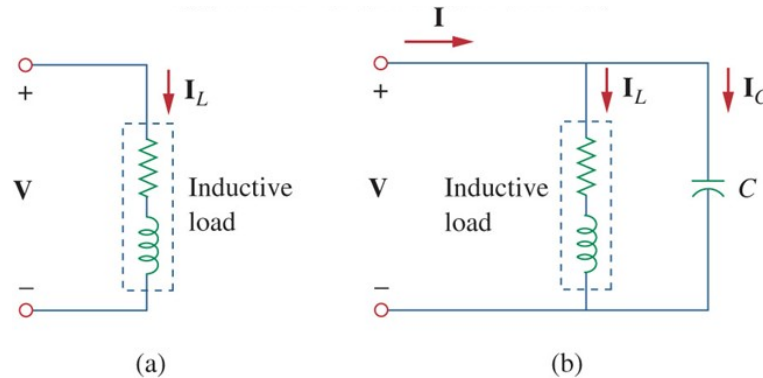
- a. Power factor
- b. Apparent power delivered to load
- c. RMS voltage

Power Factor Correction

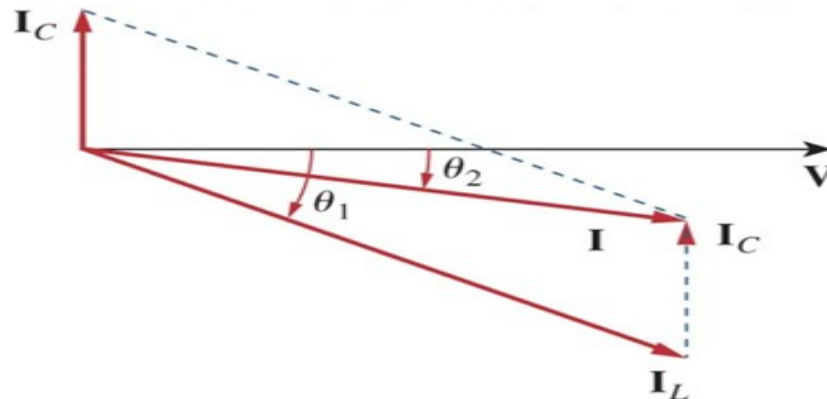
- Most domestic and industrial loads, such as washing machines, air conditioners, and induction motors are inductive => have a low, lagging power factor
- The load cannot be changed, but the power factor can be increased without altering the voltage or current to the original load
- This is referred to as power factor correction

Adding a Capacitor

- To mitigate the inductive aspect of the load, a capacitor is added in parallel with the load



- Looking at the phasor diagram, showing before and after adding the capacitor, the power factor has improved.



Adding a Capacitor

- With the same supplied voltage, the current draw is less by adding the capacitor
- Since power companies charge more for larger currents because it leads to larger power losses
- Overall, the power factor correction benefits both the power company and the consumer
- By choosing a suitable size for the capacitor, the power factor can be made to be unity

Adding a Capacitor

- The capacitor needed in order to shift the power factor angle from θ_1 to θ_2 is:

$$C = \frac{Q_c}{\omega V_{rms}^2} = \frac{P(\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

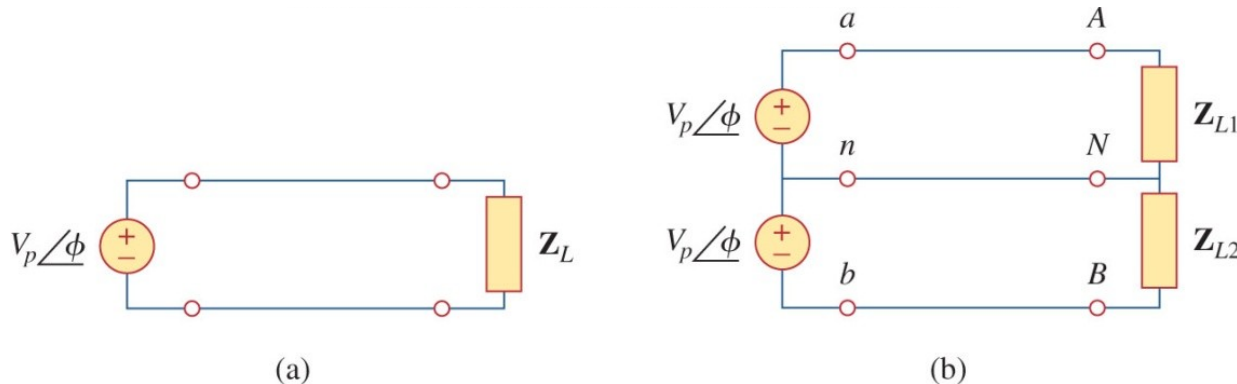
- Note that the real power dissipated in the load is not affected by the shunt capacitor

Practice Problems

11.15 Find the value of parallel capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. Assume that the load is supplied by a $110V_{\text{rms}}$, 60Hz line

Single Phase versus Polyphase

- Households have single phase mains power supplied
- This is typically done in a three wire form, where two 120V sources with the same phase are connected in series
- This allows for appliances to use either 120 or 240V



Polyphase

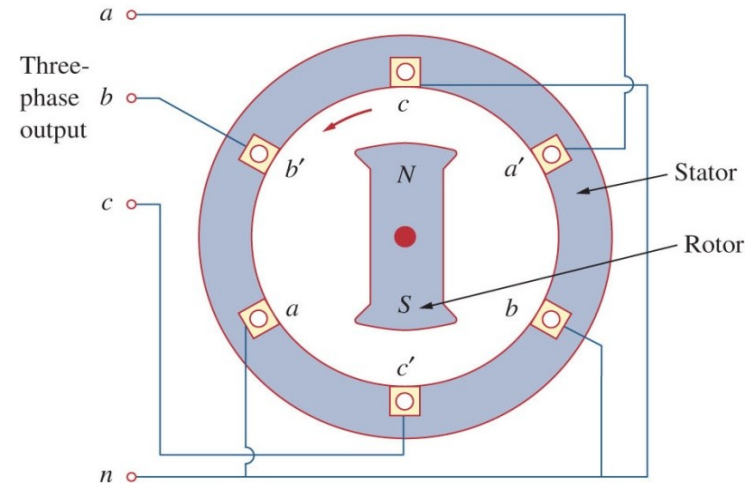
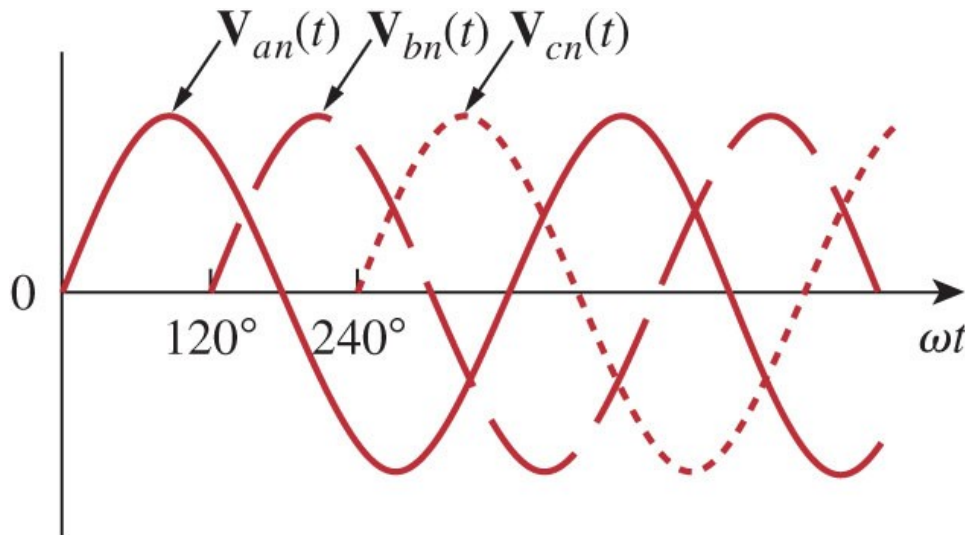
- Circuits that operate at the same frequency but with multiple sources at different phases are called polyphase
- Generating multiple phases is relatively simple when using a generator by placing coils at positions such that a lag in the current is produced leads to a phase lag
- In power grids, three phase power is used for a variety of reasons

Three Phase Power

- It is easy to extract single or two phase power from a three phase system, satisfying the cases where this is needed
- The instantaneous power in a three phase system does not pulsate like it does in a single phase system
- Lastly, the transmission of three phase is more economical than transmitting the equivalent single phase power

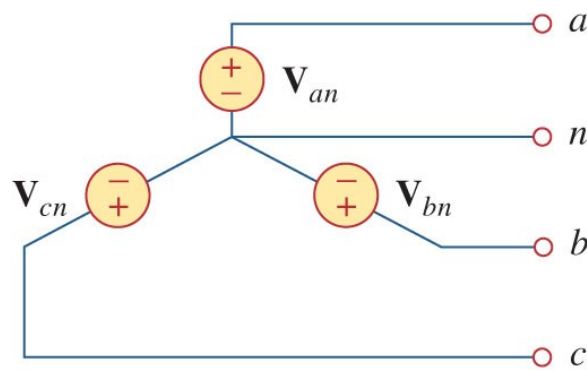
Balanced Three Phase

- Three phase voltages are typically produced by a three phase AC generator
- The output voltages look like below

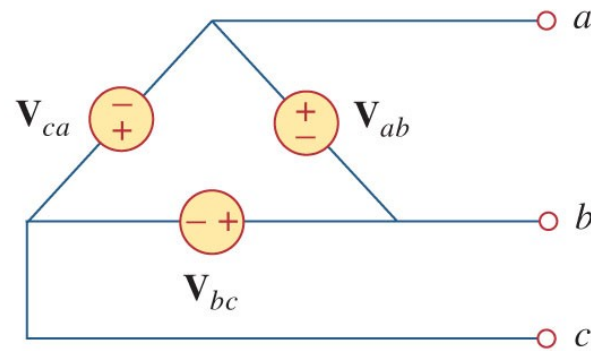


Connecting the Sources

- Three phase voltage sources can be connected to loads by either three or four wire configurations
- The three wire configuration is accomplished by a Δ connected source
- The four wire system is accomplished using a Y connected source



(a)



(b)

Balanced Source

- A Y connected source is said to be balanced when the sum of the three voltages is zero:

$$V_{an} + V_{bn} + V_{cn} = 0$$

- This can only happen if:

$$|V_{an}| = |V_{bn}| = |V_{cn}|$$

- There are two sequences for the phases:

$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ$$

$$V_{cn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

Positive sequence

$$V_{an} = V_p \angle 0^\circ$$

$$V_{cn} = V_p \angle -120^\circ$$

$$V_{bn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

Negative sequence

Phase Sequence

- The phase sequence is determined by the order in which the phasors pass through a fixed point in the phase diagram
- The phase sequence is important as it determines the direction of rotation of a motor connected to the power source
- Likewise, the phase sequence of the generator is determined by the direction in which it is turned

Balanced Loads

- Similar to the source, a balanced load is one that has the same impedance presented to all three voltage sources
- They may also be connected in either a Δ or Y configuration

- For a balanced Y connected load:

$$Z_1 = Z_2 = Z_3 = Z_Y$$

- For a balanced Δ connected load:

$$Z_a = Z_b = Z_c = Z_{\Delta}$$

Source-Load configurations

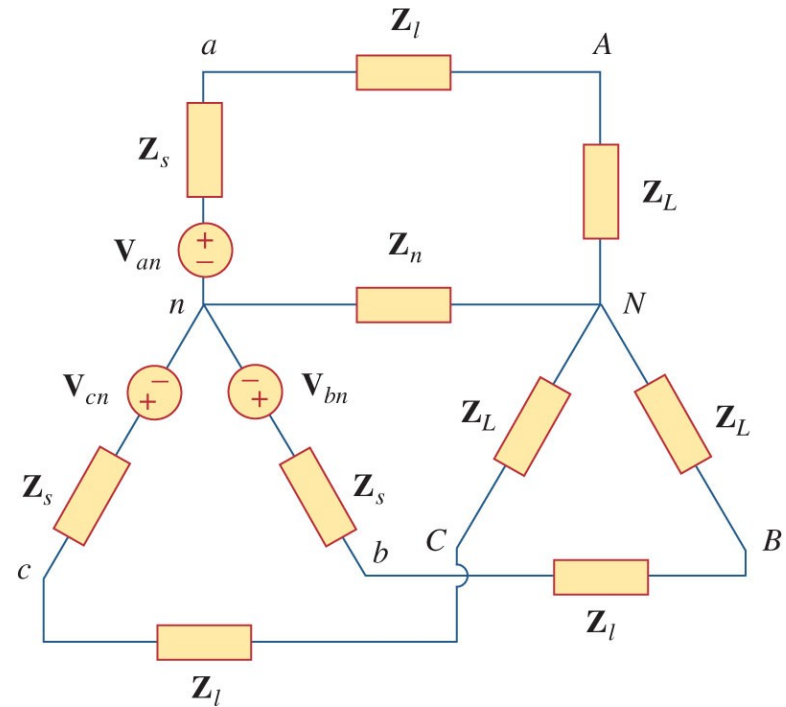
The load impedance per phase for the two load configurations can be interchanged:

This means there are four possible connections:

- Y-Y connection
- Y- Δ connection
- Δ -Y connection
- Δ - Δ connection

Balanced Y-Y connection

- Any three phase system can be reduced to an equivalent Y-Y system
- We will consider an example of a balanced four wire Y-Y system shown here
- The load impedances Z_y will be taken to be balanced
- This can be the source, line and load together

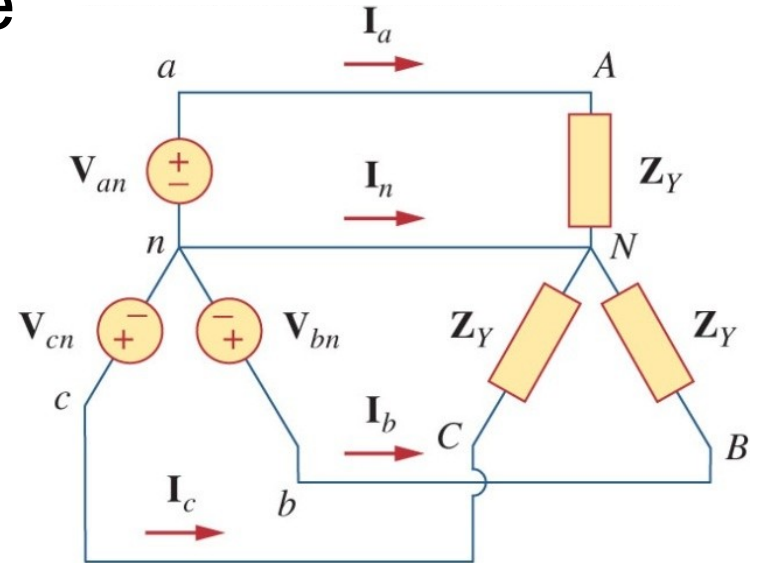


Balanced Y-Y

- Assuming the line and source impedances are small compared to the load, they can be neglected
- We will use the positive sequence for this circuit, meaning the voltages are:

$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ \quad V_{cn} = V_p \angle +120^\circ$$



Line to Line Voltages

To calculate line-to-line voltage (or line voltage):

$$\begin{aligned} V_{ab} &= V_{an} + V_{nb} = V_{an} - V_{bn} = V_p \angle 0^\circ - V_p \angle -120^\circ \\ &= V_p \left[1 + \left(\frac{1}{2} + j \frac{\sqrt{3}}{2} \right) \right] = \sqrt{3} V_p \angle 30^\circ \end{aligned}$$

Similarly,

$$V_{bc} = \sqrt{3} V_p \angle -90^\circ$$

$$V_{ca} = \sqrt{3} V_p \angle -210^\circ$$

Line to Line Voltages

- The line to line voltages are:

$$V_{ab} = \sqrt{3}V_p \angle 30^\circ$$

$$V_{bc} = \sqrt{3}V_p \angle -90^\circ$$

$$V_{ca} = \sqrt{3}V_p \angle -210^\circ$$

- Thus the magnitude of the line voltages V_L is:

$$V_L = \sqrt{3}V_p$$

- Where:

$$V_p = |V_{an}| = |V_{bn}| = |V_{cn}|$$

$$V_L = |V_{ab}| = |V_{bc}| = |V_{ca}|$$

Line Currents

- If we apply KVL to each phase, we find the line currents are:

$$I_a = \frac{V_{an}}{Z_Y} \quad I_b = I_a \angle -120^\circ$$

$$I_c = I_a \angle -240^\circ$$

- From this one can see the line currents add up to zero

$$I_a + I_b + I_c = 0$$

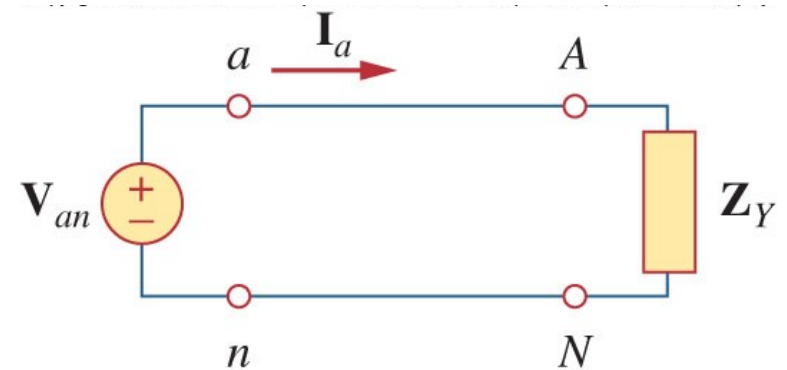
- This shows that the neutral wire has zero voltage and no current, thus it can be removed without affecting the system

Per Phase Analysis

- An alternative way to analyze the Y-Y circuit is to look at each phase individually
- Let us look at phase a
- The equivalent circuit for that phase is shown here
- The current for this phase is:

$$I_a = \frac{V_{an}}{Z_Y}$$

- If the circuit is balanced, only one phase need be analyzed

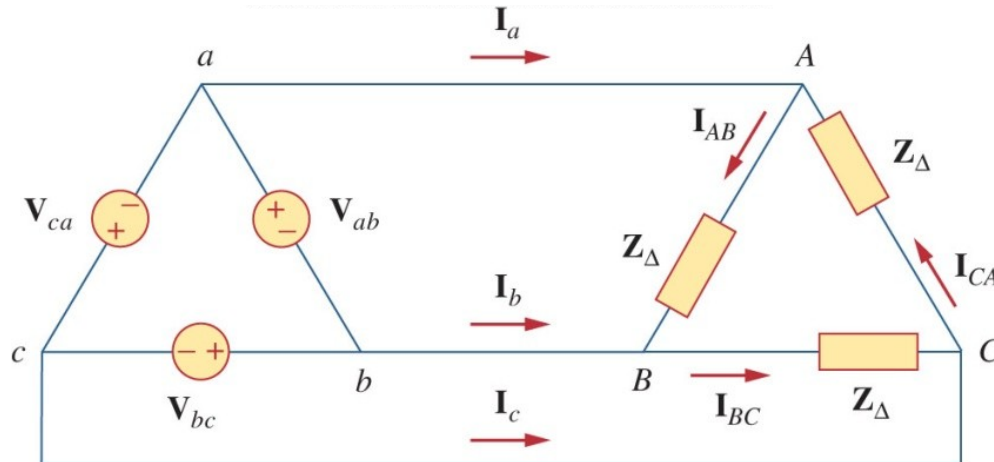


Practice Problems

- 12.1 Given that $V_{bn} = 110\angle 30^\circ$ V, find V_{an} and V_{cn} , assuming a positive (*abc*) sequence
- 12.2 A Y-connected balanced three-phase generator with an impedance of $0.4+j0.3\Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24+j19\Omega$ per phase. The line joining the generator and the load has an impedance of $0.6+j0.7\Omega$ per phase. Assuming a positive sequence for the source voltages and that $V_{an}=120\angle 30^\circ$ V, find:
- The line voltages
 - The line currents

Balanced Δ - Δ

- Now we will turn our attention to the Δ - Δ configuration
- Once again, the goal is to get the phase and line currents
- Note that Δ configured generators are less typical than the Y because any imbalance in the voltage sources will result in current flowing through the Δ loop



Convert back to Y-Y

One way to solve this system is to convert both the source and load back to a Y-Y system

- The equivalent Y connected source voltages are:

$$V_{an} = \frac{V_p}{\sqrt{3}} \angle -30^\circ \quad V_{an} = \frac{V_p}{\sqrt{3}} \angle -150^\circ \quad V_{an} = \frac{V_p}{\sqrt{3}} \angle 90^\circ$$

- The equivalent Y connected loads are found using Δ -Y transformation:

$$Z_Y = \frac{Z_\Delta}{3}$$

- With this circuit now rendered as a Y-Y circuit, single phase analysis can be done as shown before

$$I_a = \frac{V_p / \sqrt{3} \angle -30^\circ}{Z_Y}$$

Summary

TABLE 12.1

Summary of phase and line voltages/currents for balanced three-phase systems.¹

Connection	Phase voltages/currents	Line voltages/currents
Y-Y	$\mathbf{V}_{an} = V_p \angle 0^\circ$ $\mathbf{V}_{bn} = V_p \angle -120^\circ$ $\mathbf{V}_{cn} = V_p \angle +120^\circ$ Same as line currents	$\mathbf{V}_{ab} = \sqrt{3}V_p \angle 30^\circ$ $\mathbf{V}_{bc} = \mathbf{V}_{ab} \angle -120^\circ$ $\mathbf{V}_{ca} = \mathbf{V}_{ab} \angle +120^\circ$ $\mathbf{I}_a = \mathbf{V}_{an} / \mathbf{Z}_Y$ $\mathbf{I}_b = \mathbf{I}_a \angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a \angle +120^\circ$